

Hongao ZHU

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Education

Shanghai Jiao Tong University (SJTU), Shanghai, China

2021.09 – Present

B.A., English (Linguistics), Expected 2025

• Relevant Courses:

- **Linguistics:** Semantics, Syntax, Corpus Linguistics, Phonetics and Phonology, Research Methods in Linguistics, Neurolinguistics, Language and Cognition, Language and Intelligence, Language Acquisition, Hearing Science, French, Latin
- **Math and Statistics:** Calculus, Statistics for Language and Linguistic Research, Language Data and Python Techniques
- **Computer Science:** Data Structure, Natural Language Processing, Reinforcement Learning, Machine Learning, Deep Learning

Publications

Hongao ZHU, Hai HU. (in prep). Ba-BLiMP: How Does Language Models Acquire the Chinese “ba” Construction.

Yikang LIU, Hongao ZHU, Siyuan SONG, Yeting Sheng, Hai HU, Lilong XU. (in prep). ZhoBLiMP: a Systematic Assessment of Language Models with Linguistic Minimal Pairs in Chinese.

Hongao ZHU, Lei LEI. (Under Review). Detecting Authorship Between Generative AI Models and Humans: A Burrows’s Delta Approach. *Digital Scholarship in the Humanities*.

Research Experience

Ba-BLiMP: How Does Language Models Acquire the Chinese “ba” Sentences

2024.07 – Present

Undergraduate Thesis, supervised by Hai HU

Department of Linguistics, SJTU

- Formulated Ba-BLiMP, the first BLiMP(Benchmark of Linguistic Minimal Pairs) containing all paradigms of the “ba” sentences.
- Modulated Ba-BLiMP for easy “scale-up”, by presenting it with a toolchain that allows customizable grammars and vocabulary.
- Trained LMs from 14M to 12B from a carefully-curated, MinHashLSH-deduplicated dataset to cover the acquisition process.
- Manipulated the volume and diversity of “ba” sentences as input data for each checkpoint.

Language Models As Surprisal Predictors to Human Reading Times

2024.03 – Present

Research Assistant, supervised by William Schuler

Department of Linguistics, Ohio State University

- Contributed to the psycholinguistic toolkit [modelblocks](#) by adding modules for deduplication and for trimming various human-response data.
- Deduplicated the [NatrallStories](#) corpus of [Pythia](#) training data, consolidating the predictive power of small LMs in [Oh and Schuler \(2023\)](#).
- Reassessed transformer-based models’ alignment with humans on cognitive tasks and behaviors, following [Schrimpf et al. \(2021\)](#).

Detecting Authorship Between Generative AI Models and Humans: A Burrows’s Delta Approach

2023.10 – Present

Team Leader, supervised by Lei LEI

Department of Linguistics, Shanghai International Studies University

- Appropriated Burrows’s Delta, a classic stylometric algorithm, into AI detection and proved its robustness on the [ArguGPT](#) Corpus (2023).
- Implemented and tested the Delta algorithm, the SVM, and the RNN with Python, with NLP packages such as Scikit-Learn, SciPy, NumPy...
- Overrode the classic Delta implementations in Scikit-Learn for efficiency, according to the peculiar structures of data.

Multilingual Syntax Acquisition for LLMs: The case of binding and reflexives

2023.10 – Present

Research Assistant, supervised by Hai HU

Department of Linguistics, SJTU

- Encoded the binding principles across languages, allowing generations of minimal pairs in respect of finiteness, gender, and anaphor.
- Applied the benchmark to a series of multilingual models such as Qwen, eliciting performances from log probability, perplexity to prompting.

Assessing The Communication Effect of Chinese Image in Australian Public Discourse

2022.05 – 2023.09

Team Leader, supervised by Changpeng HUAN

Department of Linguistics, SJTU

- Conducted multimodal discourse analysis, source analysis, and strategy analysis on the self-built *Corpus of Chinese News from Australia*.

- Employed UCREL Semantic Analysis System and Corpus linguistics toolkits such as Wordsmith and KHCode in critical discourse analysis.

Topic Modeling And Sentiment Analysis of classical literary works across 700 years 2021.10 – 2023.09

- *Team Leader, supervised by Lei LEI* Department of Linguistics, SJTU
- Explored NLP techniques, the topic-modeling and sentiment analysis, on the Corpus of Canon of Western Literature (Green, 2017).
- Invented new stop-words lists and optimized LDA model with Perplexity calculations for historical text topic-modeling.
- Visualized the analysis in heat-maps, co-occurrence graphs, and interactive webpages with the assistance of NLTK, Matplotlib, pyLDAvis...

Workshops

University of Oxford, Oxford, The United Kingdom 2023.08 – 2023.09

Attendee, Oxford Prospects Program affiliated with Regent Park's College, 2023

- Discussed the latent negativity in discourse sentiment analysis and the question of topic segmentation with Dr Samantha-Kaye Johnston.

Hong Kong University, Hong Kong, China 2023.05 – 2023.06

Attendee, New Horizons in Digital Humanities & Cultural Data Science Workshop, 2023

- Conversed with Professor David Mimno about the applications of LDA and the feasibility of LLM-assisted essay scoring for TOFEL.

Shanghai International Studies University, Shanghai, China 2023.01 – 2023.02

Attendee, Workshop on Discourse and Multimodal Analysis, 2023

- Received lectures from prominent professors concerning multimodal and discourse analysis, including Professor David Machin.

Skills

Languages: Mandarin Chinese (Native), English (Advanced), French (Intermediate), Latin (Preliminary)

Programming Languages: Python (Scikit-learn, Scipy, PyTorch, NLTK, Numpy, Pandas, ...), R, C/C++, C#

Software: Visual Studio Code, Visual Studio, Anaconda, LaTeX, Praat, Antconc, WordSmith, KHCode, CLAN, SPSS, PS, PR

Awards and Honors

- Excellent, Test for English Majors – Band 4 (TEM-4) 2023
- Level-3, China Accreditation Test for Translators and Interpreters (CATTI) 2023
- Level-3, National Computer Rank Examination (NCRE) 2022
- First Prize, The 7th National Academic English Vocabulary Competition (NAEV) 2022
- Second Runner-up and First Prize, The 2022 China International Law of the Sea Moot Court Competition 2022
- Champion, RoboCup Small Group in Shanghai Jiao Tong University 2021

Course Projects

RL Logistics Simulator, SJTU, Shanghai, China 2024.05

- Designed a smart logistics simulator with DQN and Dijkstra, which will prevent 10% of the traffic jams in Black Friday events by prediction.

Acquisition of English Plural Marking by a Cantonese-English Bilingual Child: A Corpus-based study, SJTU, Shanghai, China 2023.09

- Presented a CHILDES-based study of simultaneous bilingual children in support of the interdependent development hypothesis.

Extracurricular Experience

Editor and Journalist, Yiyou Newspaper, SJTU, Shanghai, China 2022.09 – 2023.09

Playwright, School history drama for the 127th anniversary, SJTU, Shanghai, China 2022.02 – 2023.04

Director of the Academic Division, English Debate Club, SJTU, Shanghai, China 2021.09 – 2023.06